

Design and Development of a Multilingual Chatbot for Enhancing Accessibility in Sri Lanka's Digital Welfare Services

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Dissertation submitted in partial fulfilment of the requirements for the Bachelor of Science
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DECLARATION

I assert that this is my original work, and this research project focused on creating a Sinhala-English (SinEnglish) chatbot system does not include any material from previous submissions for a degree or diploma at any other university or higher education institution without proper acknowledgment. To the best of my knowledge and belief, it does not contain any material previously published or authored by someone else, except where appropriate credit is given in the text. This study aimed to create and deploy an intelligent chatbot that can accurately comprehend and reply to inputs in both Sinhala and English, thus enhancing communication and user engagement in bilingual settings. The system was created with contemporary technologies and approaches, guaranteeing precision, effectiveness, and ease of use. During the research process, I adhered to ethical standards and maintained academic integrity by properly citing and acknowledging all external sources, references, and contributions. All help provided during the project's development has been explicitly acknowledged, and no unauthorized material has been incorporated. I also grant the Sri Lanka Institute of Information Technology (SLIIT) the non-exclusive permission to reproduce and distribute my dissertation in full or in part in print, electronic, or any other format. I reserve the right to utilize this content, in full or in part, in upcoming works like research papers, publications, or books. Moreover, I affirm that this study adds to the realm of chatbot creation and natural language processing by tackling issues associated with multilingual comprehension and response formulation.

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Signature of the Supervisor

Date

Signature of the Co-Supervisor

Date

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In order to successfully complete my research project, "Design and Development of a Multilingual Chatbot for Enhancing Accessibility in Sri Lanka's Digital Welfare Services," I would want to sincerely thank everyone who helped and advised me.

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Thank you.

ABSTRACT

Through initiatives like Samurdhi, disability benefits, senior pensions, and nutritional support, Sri Lanka's social welfare system benefits millions of people. However, many people still find it difficult to obtain these services because of restricted internet accessibility, ignorance, and language hurdles. Communication and service delivery gaps result from current platforms' frequent inability to adequately assist customers from a variety of linguistic and educational backgrounds. The design and development of a multilingual AI-powered chatbot with the goal of improving accessibility within Sri Lanka's digital welfare services is the main emphasis of this study. With its ability to communicate with users in Sinhala, Tamil, and English, the chatbot makes it simple for citizens to learn about welfare programs, eligibility requirements, application processes, and available benefits. In order to comprehend user inquiries and deliver precise, context-aware answers instantly, the system makes use of Natural Language Processing (NLP) algorithms. To train and assess the chatbot, a structured dataset of commonly asked queries, information about welfare programs, and conversational intents was created.

Modern technologies like React.js for the frontend and AI and NLP frameworks for the backend were used to create the solution as a web-based application. Multilingual query processing, immediate results, guided assistance for welfare applications, and conversational interfaces that are easy to use for those with low levels of digital literacy are some of the system's key features. The results of this study show that a multilingual chatbot can greatly increase user engagement, accessibility, and efficiency when providing welfare-related information. The technology helps create a more inclusive and citizen-focused digital welfare ecosystem in Sri Lanka by bridging communication barriers and streamlining service access.

Keywords - Artificial Intelligence, Multilingual Chatbot, Natural Language Processing, Welfare Services, Accessibility, Digital Platforms, Sri Lanka.

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
AI	Artificial Intelligence
ANN	Artificial Neural Networks
BERT	Bidirectional Encoder Representations from Transformers
BVMI	Beery-Buktenica Developmental Test of Visual-Motor Integration
CNN	Convolutional Neural Network
DL	Deep Learning
LA	Learning Analytics
MDLS	Multidimensional Legibility Scale
ML	Machine Learning
MVC	Model, View, Controller
NLP	Natural Language Processing
POS	Part-of-Speech
SBMP	Sensory Based Motor Praxis Training
SR	Speech Recognition
SVM	Support Vector Machine
TEL	Technology-Enhanced Learning
UCD	User-Centered Design
ZPD	Zone of Proximal Development

1. INTRODUCTION

Protecting and assisting a nation's most vulnerable citizens is the main goal of social welfare programs. Through a number of government programs, including Samurdhi, the Aswesuma welfare scheme, disability payments, pensions, nutritional support programs, and housing help, Sri Lanka has fulfilled this obligation for many years. These initiatives show the government's dedication to lowering poverty, combating inequality, and making sure that people who are struggling financially, unemployed, unwell, or socially disadvantaged are not left behind.

The efficacy and accessibility of welfare programs continue to be important challenges despite these efforts and large financial investments. Due to systemic inefficiencies, ignorance, and communication hurdles, a sizable segment of the population still finds it difficult to obtain welfare assistance. Many people who qualify are either denied assistance or have trouble completing the intricate application procedures. The shortcomings of the current welfare delivery system are highlighted by this discrepancy between legislative aim and actual impact.

The conventional, manual procedures employed in the provision of social services are one of the main problems. Citizens must physically visit Divisional Secretariat offices to complete paper-based forms and turn in supporting documentation, including family records, identity cards, and income information. For people who live in remote areas, have impairments, or have limited financial and technological means, this process is time-consuming, inconvenient, and very difficult. Additionally, the decision-making process frequently relies on human judgment rather than data-driven or standardized methods, which can result in inefficiencies and discrepancies.

The language barrier that exists in Sri Lanka's multilingual setting is another significant obstacle. Many citizens have trouble comprehending welfare-related information when it is not available in their preferred language because Sinhala, Tamil, and English are commonly used. Users with low digital literacy or limited language competence find it more difficult to interact with current digital platforms since they frequently lack adequate multilingual support.

In this regard, cutting-edge technologies like artificial intelligence (AI) and natural language processing (NLP) are viable ways to increase the effectiveness and accessibility of social services. Chatbot systems have become one of these technologies' most useful tools for

facilitating interactive, real-time, and user-friendly communication. Chatbots are especially helpful for assisting users with complicated procedures since they can mimic human-like discussions, comprehend user inquiries, and deliver prompt responses. In order to improve accessibility in Sri Lanka's digital welfare services, this study focuses on the creation of a multilingual chatbot. By allowing users to communicate in Sinhala, Tamil, and English, the suggested method lowers language barriers and fosters inclusivity. It offers real-time support, responds to user inquiries, walks people through the welfare application process, and presents pertinent information in an easy-to-understand way.

The chatbot provides a conversational interface that is simple to use, especially for people with little technological expertise, in contrast to conventional static web portals. The solution guarantees that users can access welfare-related services at any time and from any location by combining AI-driven language processing with a web-based platform. By reducing the requirement for in-person visits, this strategy not only increases user involvement but also lessens the strain on government offices. The ultimate goal of this research is to show how citizens' interactions with welfare systems can be revolutionized by a multilingual chatbot. The suggested approach helps to build a more effective, inclusive, and user-centered digital welfare ecosystem in Sri Lanka by enhancing accessibility, streamlining communication, and offering immediate assistance.

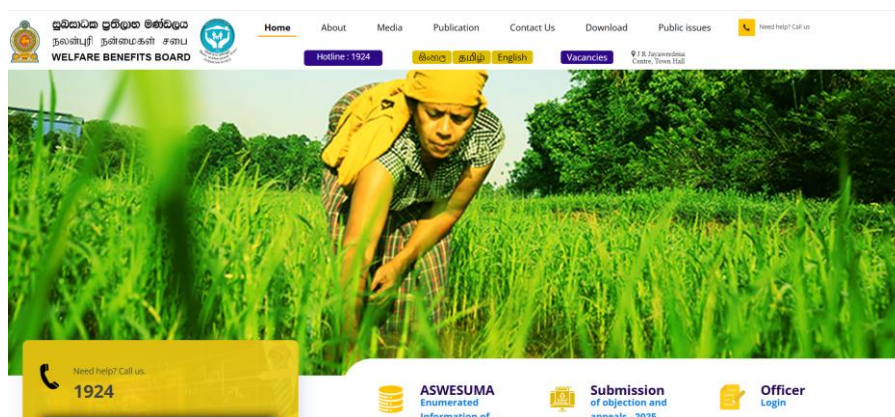


Figure 1: Current Sri Lanka Welfare Portal

The government's current digital portal for accessing welfare-related services including Samurdhi, Aswesuma, and other social assistance programs is the Sri Lanka Welfare Portal. For many citizens, the portal's usability and accessibility are still restricted, despite the fact that it offers a consolidated entry point for information and application submission.

It is clear from Figure 1's visual portrayal that the system is mostly information-heavy and form-based, requiring users to manually traverse through numerous pages and input fields. Older users, people with little computer literacy, and people unfamiliar with internet systems may find this design difficult.

Furthermore, the portal mostly functions in a static and non-interactive way, requiring visitors to read and comprehend information on their own without support in real time. Users frequently experience confusion, make wrong submissions, or visit welfare offices repeatedly because there is no conversational help to walk them through the process step-by-step.

This restriction draws attention to the discrepancy between current welfare programs and contemporary user expectations. Consequently, it is evident that an intelligent, interactive, multilingual solution is required. This study suggests a Multilingual Chatbot for Digital Welfare Services as a solution to these problems, which converts the conventional static portal into a conversational system. Users can ask queries in Sinhala, Tamil, or English and get prompt, guided answers instead of having to manually search through pages.

The chatbot acts as a virtual assistant integrated with the welfare system, helping users:

- Understand eligibility criteria easily
- Get step-by-step application guidance
- Receive instant answers in their preferred language
- Reduce dependency on physical visits and manual support

Therefore, the proposed chatbot significantly enhances the usability of the existing welfare portal by making it more interactive, accessible, and user-friendly for all citizens in Sri Lanka.

1.1 Background and Literature Survey

1.1.1 Background

Social welfare requirements and the institutional capacity to adequately address them are frequently significantly out of sync in emerging nations. In Sri Lanka, where government programs like Samurdhi, Aswesuma, disability allowances, pensions, and other social support schemes handle welfare services, this condition is very noticeable. Despite the fact that these services are designed to assist residents who are at risk, many users still find it difficult and ineffective to access them.

To apply for welfare assistance, residents must currently physically visit Divisional Secretariat offices. Along with supporting documentation such identity cards, proof of income, and household information, they have to turn in paper-based forms. Particularly for people who live in rural regions, are elderly, have restricted mobility, or have little computer literacy, this manual process is time-consuming and difficult. The effectiveness of the system is further diminished by the fact that applicants frequently do not completely understand the assistance programs for which they are qualified.

The absence of direct supervision and communication support is another significant drawback of the current welfare system. The qualifying requirements, application processes, and necessary paperwork are frequently difficult for citizens to comprehend. Furthermore, users are not guided through the procedure by an integrated system that offers bilingual help or individualized assistance. Because of this, many people rely significantly on welfare officials for information, which adds to the effort and lowers productivity.

Technology has shown great promise in recent years for enhancing the provision of public services in developing nations. Digital solutions have been effectively deployed in a number of South and Southeast Asian nations to improve welfare systems' accessibility, efficiency, and transparency. For instance, by connecting benefits to biometric identification systems, India's Direct Benefit Transfer (DBT) system has greatly increased the timeliness and accuracy of welfare payments. In a similar vein, Bangladesh has embraced internet platforms to enhance social protection services and boost the effectiveness of service delivery. However, rather than providing intelligent assistance, eligibility recommendations, or user support, the majority of

these systems primarily concentrate on financial transactions and benefit distribution.

There is a chance to change conventional welfare systems into more intelligent and user-centered platforms thanks to the quick development of artificial intelligence (AI) and machine learning (ML). In order to provide more precise and reliable decision support, AI-based systems can examine intricate patterns in citizen data, including income level, family structure, work status, health conditions, and geographic location. Machine learning models may learn from past data and gradually increase forecast accuracy, in contrast to rule-based systems.

The creation of a multilingual chatbot to improve accessibility in Sri Lanka's digital welfare services is the main emphasis of this study. By offering an intelligent conversational interface that supports Sinhala, Tamil, and English, the suggested system seeks to overcome accessibility issues and communication impediments. The chatbot is intended to help consumers by responding to questions about welfare, assisting them with the application process, and enhancing their comprehension of the services that are offered.

The technology seeks to close the gap between citizens and government services by fusing social service data with AI-driven language analysis. In addition to enhancing accessibility and user experience, this strategy helps Sri Lanka's digital welfare ecosystem become more effective, inclusive, and transparent.

1.1.2 Literature Survey

AI, NLP, and chatbot systems in e-governance and public service delivery have been the subject of a substantial amount of research. These studies demonstrate how digital technology may increase citizen engagement with government services, decrease administrative burden, and improve accessibility.

Følstad and Brandtzaeg (2017) discovered that by enabling individuals to converse using natural language rather than complicated online navigation, chatbots enhance user participation in public services. Services become more accessible as a result, particularly for users with low levels of digital literacy. After reviewing chatbot technologies, Adamopoulou and Moussiades (2020) came to the conclusion that AI-based chatbots that use natural language processing are more adaptable and scalable than straightforward rule-based systems. Nevertheless, a lot of current technologies are not tailored for the welfare or government sectors.

According to Ahmed's (2019) research on AI in social welfare systems, machine learning outperforms manual techniques in terms of eligibility prediction accuracy. However, without any justification or user advice, the majority of systems just offer simple decisions (eligible/not eligible).

Language obstacles, limited digital literacy, and resistance to digital adoption are some of the main e-governance issues that developing nations face, according to Bhatnagar (2004). These concerns are extremely pertinent to the welfare system in Sri Lanka.

The significance of multilingual NLP systems in enhancing accessibility was highlighted by Kumar et al. (2021). They discovered that providing support for regional languages greatly boosts system usage and user happiness, particularly among rural people.

Although internet availability is growing in Sri Lanka, digital literacy is still uneven, according to a 2019 World Bank assessment. Citizens find it challenging to use online government services as a result. According to Gefen et al. (2003), open communication and transparency are key components of user confidence in digital systems. To foster trust in welfare systems, people require straightforward, intelligible, and trustworthy responses. All things considered, current research indicates that chatbots and AI technology can enhance the provision of public services. Systems that integrate welfare advice, multilingual assistance, and citizen-friendly communication on a single platform are still lacking, nevertheless.

By creating a multilingual chatbot for Sri Lanka's digital welfare services, this research fills that gap and enhances user assistance, accessibility, and communication.

1.2 Research Gap

An area within a study domain that has not been adequately investigated, addressed, or resolved in prior research is referred to as a research gap. It draws attention to chances for fresh approaches, inventions, or advancements that can overcome current constraints in systems, practices, or knowledge.

There are a number of gaps between the potential provided by contemporary Artificial Intelligence (AI) and Natural Language Processing (NLP) technology and the social welfare system in Sri Lanka. Even though welfare programs like Samurdhi and Aswesuma give needy communities vital financial help, the methods utilized to provide these services are still primarily manual, disjointed, and non-interactive.

The absence of intelligent, multilingual conversational systems in the welfare sector is one of the significant research gaps found. The lack of real-time conversational support in Sinhala, Tamil, and English on current digital platforms restricts accessibility for users with varying linguistic preferences. Because of this, many residents have trouble comprehending the requirements for eligibility, the application process, and the necessary paperwork.

The existing welfare systems' lack of automated guiding and assistance tools is another major flaw. The majority of current platforms only offer static data or rudimentary options for submitting applications. They don't provide intelligent support like interactive eligibility clarification, document verification aid, or step-by-step application coaching.

Additionally, welfare service databases and AI-driven technologies are not well integrated. Citizen data, such as income level, family history, or current welfare status, is not efficiently used by current systems to deliver intelligent recommendations or tailored responses. As a result, service delivery is less accurate and efficient.

The literature also reveals a deficiency in addressing issues related to digital inclusion and accessibility, especially for rural and low-income communities. Many of the AI-based solutions that are now in use require a certain degree of computer literacy, which is impractical in Sri Lanka. Users who are older or have little experience with technology frequently find it difficult to engage with digital platforms.

Furthermore, the current welfare infrastructure lacks citizen-centric, round-the-clock support mechanisms. Convenience is diminished and service access is delayed since citizens are still forced to rely on physical offices or limited working-hour assistance.

The lack of attention paid to multilingual natural language processing for regional languages like Tamil and Sinhala is another significant gap. Since many current AI systems are made for international languages like English, they are less useful in local government service settings. These shortcomings make it clear that Sri Lanka's welfare system needs a more sophisticated, clever, and user-friendly solution that can remove obstacles to communication, increase accessibility, and offer help in real time.

Feature	This Research	Existing Systems
Multilingual Support	Yes	No
Real-time Application Tracking	Yes	Partial
Automated Document & Eligibility Assistance	Yes	No
Conversational AI for Inclusivity	Yes	No
24/7 Citizen Support	Yes	No
Integration with Welfare Databases	Yes	Limited

Table 1: Comparison of This Research with Existing Systems

1.3 Research Problem

How to create and deploy an AI-powered multilingual chatbot system that can efficiently increase accessibility to Sri Lanka's welfare services while offering citizens accurate, timely, and user-friendly support is the main research question this project attempts to solve. The system must be able to comprehend user inquiries, handle different languages (English, Tamil, and Sinhala), and provide insightful answers that help users navigate welfare-related procedures.

Citizens of Sri Lanka confront a number of difficulties with the country's present welfare system, including poor computer literacy, language barriers, a lack of real-time help, and trouble comprehending eligibility requirements and application processes. The majority of current systems are static and lack interactive assistance, which makes it challenging for users to effectively utilize services. An intelligent conversational system that can close this gap is therefore obviously needed.

- How can a multilingual chatbot take into account the linguistic and cultural diversity of Sri Lankan users while accurately processing and responding in Sinhala, Tamil, and English?
- How can intent detection, entity recognition, and multilingual dialogue management be efficiently handled in a single system using Natural Language Processing (NLP) techniques?
- How can the chatbot deliver pertinent, accurate, and timely welfare information in an easy-to-understand way, such as eligibility requirements, application processes, and necessary documents?
- How can the system guarantee usability and accessibility for citizens with little technical expertise and low digital literacy?
- How can the chatbot be made to provide dependable, context-aware, and consistent responses that increase user happiness and trust?
- How can the chatbot's performance be regularly assessed and enhanced across various user groups and languages?

These difficulties are social and usability-focused in addition to technological. They include concerns about user experience design, language inclusion, fairness, and accessibility. Additionally, when citizens interact with welfare services, the system must make sure that information is presented in a way that is both practically valuable and easy to grasp.

2. Objectives

2.1 Research Objective

This study's primary goal is to create an AI-powered multilingual chatbot system that offers precise, instantaneous, and user-friendly assistance in Sinhala, Tamil, and English, thereby increasing accessibility and efficiency in Sri Lanka's social services. By providing interactive assistance with eligibility information, application processes, necessary documentation, and welfare program recommendations, the system seeks to improve communication between citizens and welfare institutions.

The goal of this research is to establish a more responsive and inclusive welfare assistance system by combining Natural Language Processing (NLP), multilingual dialogue management, and user-centered design into a single digital platform.

2.2 Specific Objectives

- To gather and preprocess a pertinent dataset of welfare-related inquiries and citizen data, including spelling errors, linguistic variances, and real-world communication patterns like unclear or insufficient user input.
- To create and apply multilingual Natural Language Processing (NLP) models that can handle entity recognition, intent detection, and dialogue management in Sinhala, Tamil, and English while taking Sri Lankan users' linguistic and cultural variations into account.
- To provide an interactive chatbot interface that facilitates text-based communication and is optimized for users with different levels of digital literacy, guaranteeing usability, clarity, and simplicity.
- In order to give precise and timely answers regarding eligibility requirements, application processes, document requirements, and available welfare programs, the chatbot will be integrated with current welfare information sources.
- To improve user experience and confidence in digital welfare services by making sure the system provides consistent, dependable, and context-aware responses in all supported languages.
- To put in place a framework for ongoing assessment and development of the chatbot based on user input, performance indicators, and interaction data in order to gradually improve accuracy, responsiveness, and overall system efficacy.

3. METHODOLOGY

3.1 Methodology Description

The design, development, and evaluation of the Multilingual Chatbot for Improving Accessibility in Sri Lanka's Digital Welfare Services are described in this section. The paradigm is realistic and iterative, emphasizing multilingual communication, accessibility, and real-world usability over theoretical performance.

Data collecting and preprocessing, chatbot model development, system integration, and interface design and deployment were the four primary stages of the development process. Every stage was constructed gradually, with each stage's production having a direct impact on the subsequent one.

The first stage involved gathering and organizing pertinent information on welfare services, user inquiries, and multilingual expressions in Sinhala, Tamil, and English. Preprocessing activities like text cleaning, managing user input variations, and creating intent-based training data appropriate for NLP processing were also included in this step.

Using Natural Language Processing (NLP) methods, the second stage concentrated on creating the chatbot's fundamental intelligence. In order to guarantee that the chatbot can comprehend user inquiries and deliver precise, context-aware responses in a variety of languages, this included intent categorization, entity recognition, and response generation techniques.

The final stage involved connecting the chatbot system to organized welfare data and integrating it into a backend architecture. This made it possible for the chatbot to provide real-time answers about eligibility, application processes, necessary paperwork, and general welfare advice.

Designing and implementing the user interface was the last stage. To guarantee accessibility for users with varying degrees of digital literacy, a straightforward, user-friendly web-based chat interface was created. To increase usability, response clarity, and multilingual performance, iterative changes were implemented based on testing and feedback.

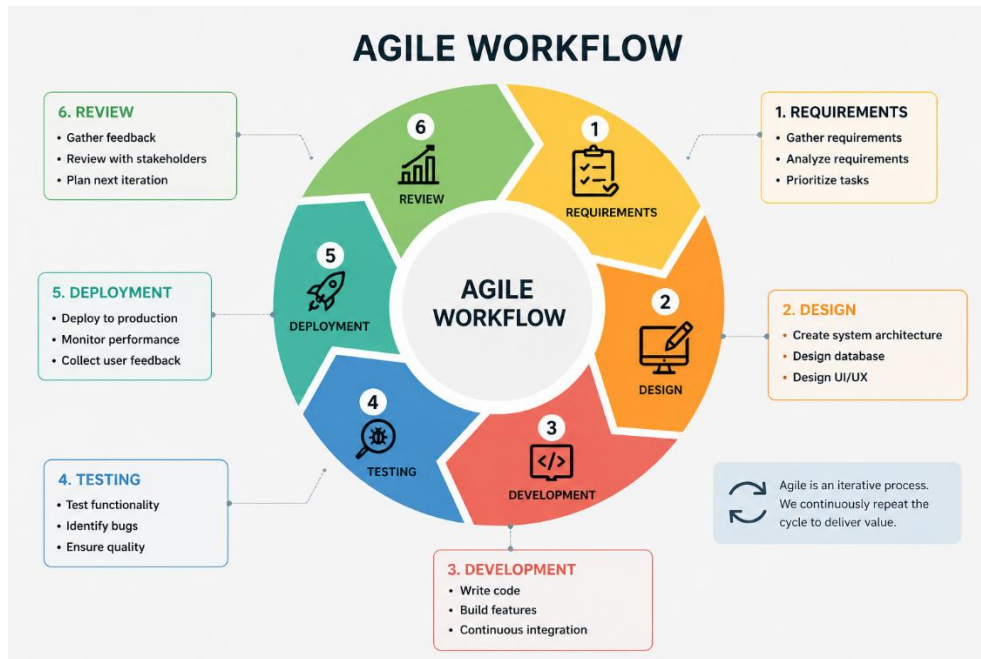


Figure 3.1: Agile workflow

The iterative development method used in the creation and deployment of the Multilingual Chatbot for Improving Accessibility in Sri Lanka's Digital Welfare Services is represented by the Agile workflow shown in Figure 1. This approach was selected to guarantee adaptability, ongoing enhancement, and user-centered development during the course of the project.

The first step in the process is requirements gathering, which identifies system needs like user accessibility requirements, welfare domain knowledge, and multilingual support (Sinhala, Tamil, and English). This step guarantees that the chatbot tackles actual user issues like language limitations, a lack of direction, and trouble obtaining welfare information.

The general system architecture, chatbot flow, and user interaction model were created during the design phase. A straightforward and user-friendly interface that is appropriate for people with low levels of digital literacy was given special consideration. Conversation flow planning and the incorporation of Natural Language Processing (NLP) elements were also included in the design.

Using AI and NLP techniques to create the chatbot was the main emphasis of the development phase. This covers multilingual response generation, entity extraction, and intent identification. The chatbot was trained to answer typical welfare-related questions in all supported languages with precision and context awareness.

In order to verify accuracy, response quality, and language handling capability, the system was tested using sample user inputs. To enhance system performance, conversational flow errors and misunderstandings were found and fixed.

During the deployment phase, the chatbot was integrated into a web-based platform so that welfare officers and citizens could access it. After that, the system was made accessible for practical use, enabling ongoing communication and feedback gathering.

Lastly, by gathering user input and improving chatbot responses, the Review and Feedback process guarantees ongoing development. Over time, the system becomes more accurate, dependable, and user-friendly because to this iterative cycle, which enables it to change based on actual user interactions.

3.2 System Overview

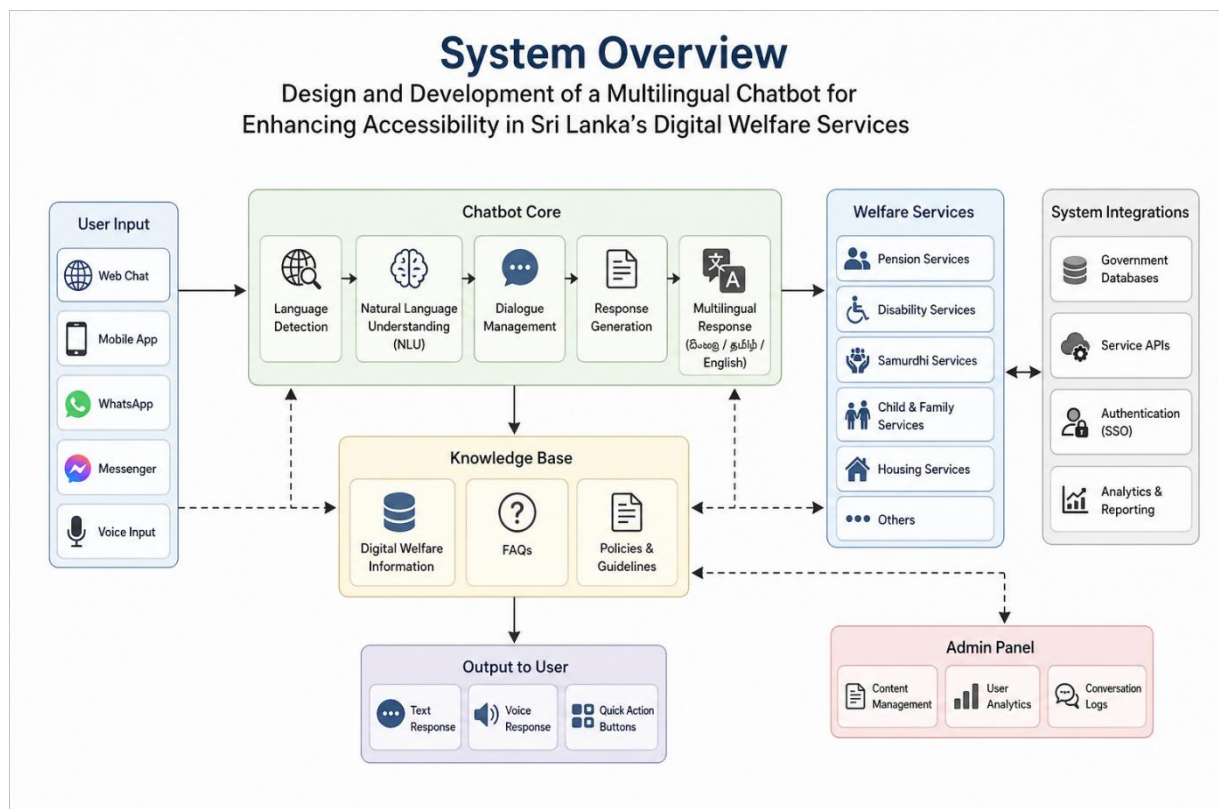


Figure 3.2: Overview of diagram

The full system overview of the suggested AI-powered multilingual chatbot for improving accessibility in Sri Lanka's digital welfare services is shown in Figure 2.2. In order to guarantee accessibility, scalability, and effective multilingual communication, the diagram shows all of the system's main parts and how they interact.

Users can communicate with government welfare services in Sinhala, Tamil, and English thanks to the system's development, which makes use of artificial intelligence (AI), natural language processing (NLP), and system integration techniques. The system's primary goal is to give citizens using digital welfare services precise, timely, and easy-to-use support.

The government social service infrastructure and end users (citizens) are the system's core stakeholders, as seen in Figure 2.2. Through a variety of interfaces, including web apps, mobile apps, voice input, and messaging services, users engage with the system. This guarantees inclusion for users with varying degrees of technology access and digital literacy.

The backend architecture of the system incorporates several advanced technologies, including:

- Natural Language Processing (NLP) techniques
- Machine Learning (ML) models for intent classification and response generation
- RESTful APIs for system communication and data exchange
- Preprocessing and language detection modules
- Dialogue management systems for maintaining conversational context
- Government database and service integrations

Interpreting user inquiries about welfare services, such as eligibility verification, application tracking, document guidance, and service information, is the system's primary function. Using integrated knowledge bases and government data sources, the chatbot interprets user input, recognizes purpose and entities, and produces suitable multilingual responses.

3.3 Requirements

3.3.1 Functional requirement

- **User Authentication (Secure Login System)**

The system shall allow users (citizens, administrators, and officers) to securely log in using their credentials to ensure authorized access to the platform.

- **Multilingual Chatbot Interaction**

The system shall provide an AI-powered chatbot interface that supports Sinhala, Tamil, and English, enabling users to interact in their preferred language.

- **Welfare Information Retrieval**

The system shall allow users to request and receive information about available welfare schemes, eligibility criteria, and application procedures through chatbot queries.

- **Eligibility Prediction and Recommendation**

The system shall analyze user-provided data and predict eligibility for welfare programs, while also providing suitable recommendations based on user profiles.

- **Guided Application Assistance**

The system shall assist users step-by-step in completing welfare application processes through conversational guidance provided by the chatbot.

- **Natural Language Query Processing**

The system shall process user inputs using NLP techniques to understand intent, extract relevant information, and generate accurate responses.

- **User Profile Management**

The system shall allow users to create, view, and update their personal and socioeconomic profiles, which are used for generating recommendations and eligibility analysis.

- **Real-time Response Generation**

The system shall provide real-time responses to user queries through the chatbot, ensuring efficient and interactive communication.

- **Feedback and Interaction Tracking**

The system shall record user interactions and feedback to improve chatbot performance and enhance future responses.

3.3.2 Use cases diagram

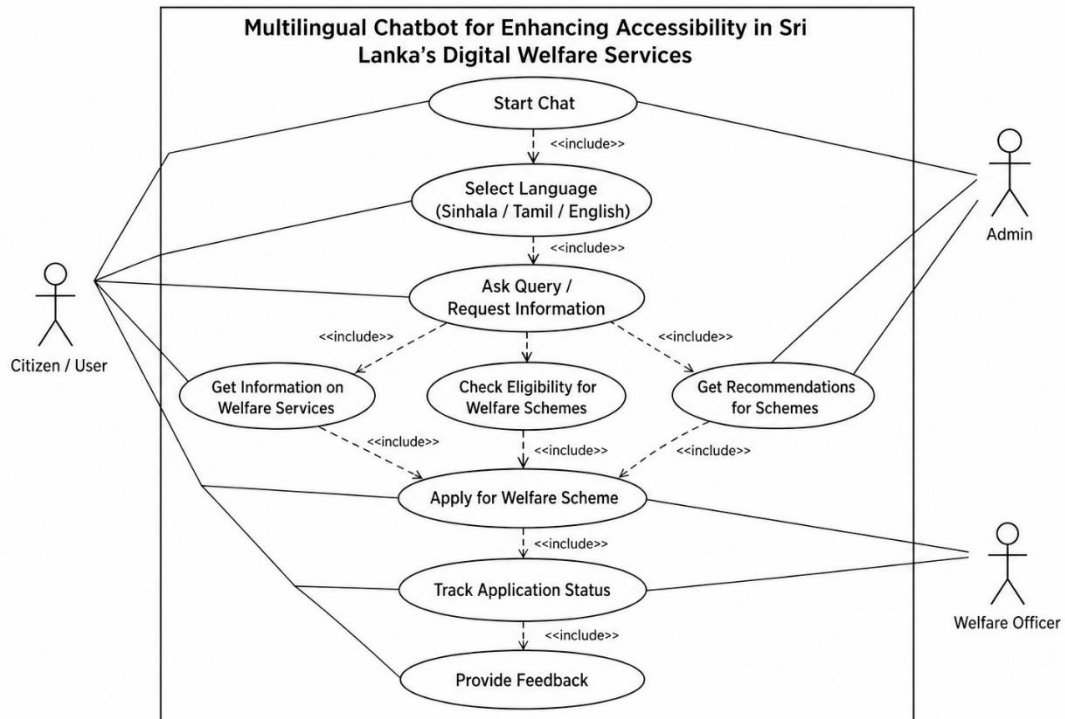


Figure 3.3.2 : Use cases diagram

3.3.3 System use cases scenarios

Use Case 1: User Queries Welfare Information

Description:

A citizen asks general information about available welfare programs.

Main Flow:

1. User sends a query in Sinhala / Tamil / English.
2. Chatbot detects language automatically.
3. NLP engine interprets user intent.
4. System retrieves relevant welfare program details.
5. Chatbot responds in user's language.

Outcome:

User receives clear information about available welfare schemes.

Use Case 2: Eligibility Checking for Welfare Programs

Description:

User checks whether they qualify for a specific welfare program.

Main Flow:

1. User requests eligibility check.
2. Chatbot asks required details (income, family size, etc.).
3. User provides information via text/voice.
4. System processes data using eligibility engine.
5. Result is calculated from welfare database rules.
6. Chatbot returns eligibility status.

Outcome:

User receives instant eligibility result.

Use Case 3: Application Status Tracking

Description:

User checks the status of an ongoing welfare application.

Main Flow:

1. User enters application ID or NIC number.
2. Chatbot validates input.
3. System connects to government welfare database.
4. Retrieves current application status.
5. Chatbot displays status update (approved / pending / rejected).

Outcome:

User gets real-time application status.

Use Case 4: Document Submission Guidance

Description:

User requests help regarding required documents for applications.

Main Flow:

- 6 User asks about document requirements.

- 7 NLP engine identifies program type.
- 8 System fetches required document list.
- 9 Chatbot explains steps clearly in user's language.
- 10 Optional: provides downloadable instructions.

Outcome:

User understands required documents and submission process.

3.3.5 Non-functional requirement

- **Reliability**

Inquiries about welfare should be consistently and accurately answered by the chatbot system. Particularly during crucial operations like eligibility verification, application tracking, and real-time information retrieval from government systems, it must function flawlessly.

- **Scalability**

Over time, the system should be able to accommodate an increasing number of users. The chatbot design must be able to scale well to manage growing traffic without experiencing performance deterioration as more residents use digital welfare services.

- **Performance**

User input should be processed by the system with the least amount of latency. To guarantee seamless and quick user engagement, natural language processing (NLP), intent identification, and data retrieval from welfare databases should be carried out in real time.

- **Security**

Strong user data protection, especially sensitive and private welfare information, must be provided by the system. To stop illegal access to user and government data, secure authentication, encrypted communication, and access control systems should be put in place.

- **Availability**

There should be less downtime and the chatbot system should be accessible around-the-clock. Users should have uninterrupted access to welfare services via message, mobile, and web platforms at all times.

- **Maintainability**

Updating and maintaining the system should be simple. Without compromising current functionality, new additions, bug corrections, language model enhancements, and interaction with other welfare programs ought to be feasible.

- **Usability**

The interface of the chatbot should be straightforward, easy to use, and intuitive. Clear directions, simple navigation, and multilingual support in Sinhala, Tamil, and English should help users with low levels of digital literacy.

- **Accuracy**

The system must guarantee a high degree of accuracy in comprehending user inquiries and delivering accurate answers. To decrease misunderstandings and increase response accuracy, NLP models ought to be trained on welfare-specific information.

3.3.7 Challenges

Obstacles Faced and Overcome

A number of difficulties arose during the creation of the Multilingual AI Welfare Chatbot for Sri Lanka, necessitating attentive problem-solving and adaption. The main obstacles that were encountered and effectively overcome are listed below.

Managing Sinhala, Tamil, and English in one system was one of the main issues. Accurate NLP model training was challenging due to the dearth of structured datasets for Tamil and Sinhala. Furthermore, distinct preparation methods and language-specific handling were needed due to the grammatical variations across the three languages. Multiple datasets were gathered, cleaned, and optimized in order to get around issue, and different language models were used to increase accuracy.

Another major problem was integrating the chatbot with the government's current welfare databases. Real-time API integration was not intended for many of these platforms at first. This made it challenging to obtain real-time data, including eligibility details and application status. In order to enable seamless system-to-system communication, middleware solutions and structured data mapping techniques were established.

Backend processing delays made it difficult to guarantee real-time responses for user inquiries like eligibility checks and application tracking. Response times were periodically impacted by slow database queries and system load. To increase performance, optimization strategies like caching, query optimization, and effective data retrieval techniques were used.

Different languages, typing techniques, and colloquial idioms are used by users to communicate with the chatbot. The NLP engine had trouble identifying intent because of this unpredictability. Text normalization, spell correction, and enhanced intent classification models were used to overcome this problem and better comprehend user inputs in the real world.

Scalability became a major challenge because the system is supposed to service a huge number of citizens. Increased user load may cause performance degradation in the absence of appropriate design. In order to get around this, a scalable architecture with optimized backend services was created to guarantee steady performance even in situations with heavy traffic.

Data security was crucial since the system manages sensitive personal data, including NIC numbers, income records, and welfare eligibility information. Strong security measures were necessary to safeguard user privacy and stop illegal access. Controlled access regulations, encrypted data transmission, and authentication methods were used to address this.

It was difficult to test the chatbot in three different languages and in a variety of real-world situations. Unexpected inputs and variations in user requests necessitated ongoing debugging and improvement. To progressively increase the correctness and dependability of the system, an iterative testing strategy was used.

3.4 Commercialization Aspects of the Product

A digital tool called the AI Multilingual Welfare Chatbot for Sri Lanka was created to increase government welfare services' responsiveness, accessibility, and transparency. The technology facilitates simple interaction with welfare programs like eligibility verification, application tracking, and document guidance by supporting citizens in Sinhala, Tamil, and English. This chatbot offers a centralized intelligent system that combines Natural Language Processing (NLP), real-time government data access, and user-friendly contact channels, in contrast to current manual or fragmented service delivery techniques. This makes it a special and useful way to enhance Sri Lanka's delivery of digital public services.

As of right now, Sri Lanka's welfare sector lacks a fully functional multilingual AI-based chatbot system. As a result, this solution offers great potential for institutional deployment, government adoption, and the digital transformation of public services. The system's practicality and efficacy have been developed based on the actual needs of welfare users and the demands of digital inclusion. The system is in line with increasing access for multilingual people, rural communities, and those with low levels of digital literacy.

The proposed chatbot system has strong **commercialization potential** in the following areas:

- **Government Integration:** It can be used to automate citizen assistance services in ministries and welfare departments.
- **Public Sector Digital Transformation:** By lowering manual labor and increasing service effectiveness, it supports Sri Lanka's e-government programs.
- **Institutional Licensing:** Adaptable and licensed for use in other government departments, including labor services, health, and education.
- **Scalable Deployment:** For broader citizen access, it can be expanded to web platforms, SMS-based systems, and mobile apps.

By verifying the platform's efficiency, accuracy, and scalability, specialists in digital transformation and public sector systems can further improve its legitimacy and usability. Their input would aid in ensuring compliance with both practical administrative requirements and national digital governance frameworks.

3.4.1 Target market and pricing model

To guarantee broad accessibility, scalability, and long-term sustainability, the AI Multilingual Welfare Chatbot for Sri Lanka employs a government-oriented deployment and service approach. The concept is intended to be a public digital service platform that can be included into the country's welfare infrastructure, rather than a conventional commercial pricing structure.

Plan	Description	Access Model
Citizen Access Platform	Gives all people free access to welfare services like document advice, application tracking, and eligibility verification.	Free Public Access
Government Institution Plan	An integrated system that reduces manual labor and automates citizen support services for ministries and welfare departments.	Government Subscription / Integration
Educational & Awareness Plan	Utilized in public awareness campaigns and digital literacy initiatives to inform people about social services.	Free / Supported Deployment
Extended Services Plan	Sophisticated reporting and analytics tools for government decision-making and policy reform.	Institutional Licensing

Table 2: Chatbot Deployment Plans and Service Model

A digital-first adoption strategy will be used to launch the AI Multilingual Welfare Chatbot through government communication channels, Facebook, and YouTube. These platforms will be utilized to increase citizen involvement, showcase chatbot functionality, and raise awareness of welfare programs.

In order to boost adoption, cooperation will be formed with:

- Departments of government welfare
- Institutions of public service
- Digital transformation initiatives in Sri Lanka

To demonstrate the efficacy of the system and foster user trust, pilot deployments and free demos may be carried out in specific areas. Long-term scalability and sustainability will also be supported by integration into digital service platforms and national e-government initiatives. Adoption will be further boosted by citizen awareness campaigns and government staff training sessions.

3.5 Testing and Implementation

3.5.1 Implementation

The AI Multilingual Welfare Chatbot for Sri Lanka is a web-based intelligent technology intended to increase government welfare services' efficiency and accessibility. To guarantee seamless connection between users and backend services, the system incorporates a number of technologies.

React.js is used in the development of the application's frontend, allowing for a user-friendly and responsive interface that is compatible with Sinhala, Tamil, and English. This makes it simple for consumers to engage with welfare services including document advice, application monitoring, and eligibility verification.

Python is used in the development of the backend, which manages request processing, communication between the frontend and database systems, and the fundamental business logic. High-performance APIs that facilitate quick and effective data interchange between system components are made with FastAPI.

User data, welfare application data, and system logs are stored in a database system (MongoDB). This guarantees scalability and effective data retrieval when managing a big user base. A Natural Language Processing (NLP) engine, which handles user questions in

Sinhala, Tamil, and English, powers the system's basic intelligence. To correctly comprehend user requests, it carries out operations including language detection, intent identification, and entity extraction.

Because the system is implemented as a real-time web application, consumers can utilize web browsers to access welfare services at any time. Future improvements and interaction with other government services are made possible by the architecture's scalability, security, and maintainability.

3.5.2 Testing

The AI Multilingual Welfare Chatbot underwent extensive testing to guarantee its accuracy and dependability. To verify system performance across various modules, both functional and integration testing were carried out.

Test cases were designed to evaluate key functionalities such as:

- Welfare information retrieval
- Eligibility checking
- Application status tracking
- Document guidance
- Multilingual response generation

The accuracy of the chatbot's responses was confirmed by comparing them to the anticipated results. To ensure robustness, the system was further tested for handling mixed-language statements, faulty inputs, and unclear inquiries.

To assess response time and system effectiveness under numerous user queries, performance testing was carried out. The system generated responses with little latency while maintaining acceptable performance levels. Security testing ensured that sensitive user data such as NIC numbers and personal welfare details were handled securely using proper authentication and access control mechanisms.

3.5.3 Functional Testing

- **Unit Testing:** To guarantee proper operation, each chatbot module is tested independently. This comprises the eligibility engine, response generation system, application tracking module, and NLP processing (language detection, intent recognition). Prior to integration, each component is independently checked.
- **Integration Testing:** To guarantee efficient communication and appropriate data flow between modules, all components, including the frontend (React interface), backend (Python APIs), NLP engine, and welfare database, are combined and tested collectively.
- **System Testing:** The chatbot system is tested as a whole. End-to-end processes like user query → language identification → intent processing → database retrieval → response creation are included in this. The accuracy of the system is assessed in welfare-related tasks including tracking applications and verifying eligibility.
- **Regression testing** is done following updates, bug patches, or feature enhancements to make sure that current features (such database integration, eligibility systems, and multilingual replies) continue to function properly without interfering with earlier ones.

3.5.4 Non-Functional Testing

- **Usability Testing:** To make that the chatbot's interface is easy to use and accessible for all citizens, even those with limited digital literacy, it is tested. Clear navigation, comprehensible responses for welfare services, and multilingual support (Sinhala, Tamil, and English) are given special consideration.
- **Security testing** guarantees the safe storage and transmission of sensitive user data, including NIC numbers, income details, and application information. Authentication mechanisms, role-based access control, and data encryption are implemented to prevent unauthorized access.

- **Performance Testing:** To guarantee real-time engagement, the system's efficiency and response time are evaluated. It assesses the speed at which the chatbot responds to user inquiries and retrieves information from government databases.
- **Compatibility Testing:** The chatbot is tested on various platforms and devices, including tablets, smartphones, and desktop browsers. It guarantees that multilingual information is shown correctly and performs consistently in all supported settings.

3.5.5 Gantt Chart

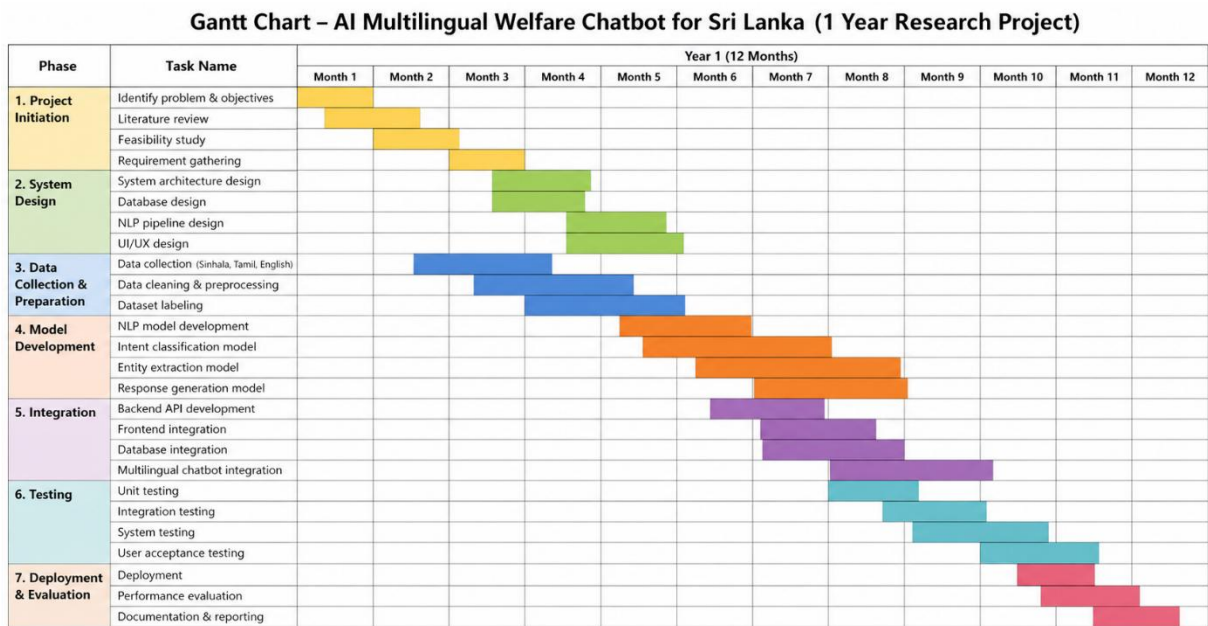


Figure 3.5.5 : Gantt Chart

4. RESULT AND DISCUSSION

4.1 Results

The developed system, "Design and Development of a Multilingual Chatbot for Enhancing Accessibility in Sri Lanka's Digital Welfare Services," was put into use as a web-based conversational application with the goal of increasing the efficiency and accessibility of retrieving welfare-related data. To meet the linguistic diversity of Sri Lanka's people, the system was built to support various languages, specifically Sinhala, Tamil, and English. By offering an intelligent, automated response system, the system's main objective was to close the communication gap between citizens and digital government welfare services.

In order to comprehend user inquiries and provide insightful answers, the chatbot was created utilizing Natural Language Processing (NLP) techniques in conjunction with an intent classification model. To train the algorithm, a structured dataset of welfare-related queries and responses was created. User intent, including eligibility verification, document requirements, application processes, and service availability, could be recognized by the system. The chatbot retrieved the best response from the knowledge base after determining the purpose.

The chatbot performed well in answering commonly asked queries during testing. The majority of the time, the system was able to accurately decipher user input and deliver pertinent answers quickly. Compared to traditional manual query methods, where consumers frequently encounter delays or uncertainty when visiting government websites, this greatly enhanced user experience.

Accessibility was greatly enhanced by the bilingual option. The accuracy of responses was unaffected by users interacting in Sinhala, Tamil, or English. Users with varying educational and language backgrounds were guaranteed inclusion thanks to this function. Additionally, the technology enhanced flexibility and usefulness by enabling simple language switching during communication.

Nevertheless, during testing, certain restrictions were noted. The chatbot occasionally failed to accurately determine the intent when user input was complicated, ambiguous, or grammatically incorrect. Due to the scarcity of training data, this was more prevalent in Sinhala and Tamil inputs. In these situations, the system either asked the user for clarification or gave a generic response.

The outcomes show that the chatbot's primary goal of enhancing access to welfare-related information was effectively accomplished. The system demonstrated the efficacy of NLP-based conversational systems in digital public service environments, greatly decreased response time, and enhanced accessibility.

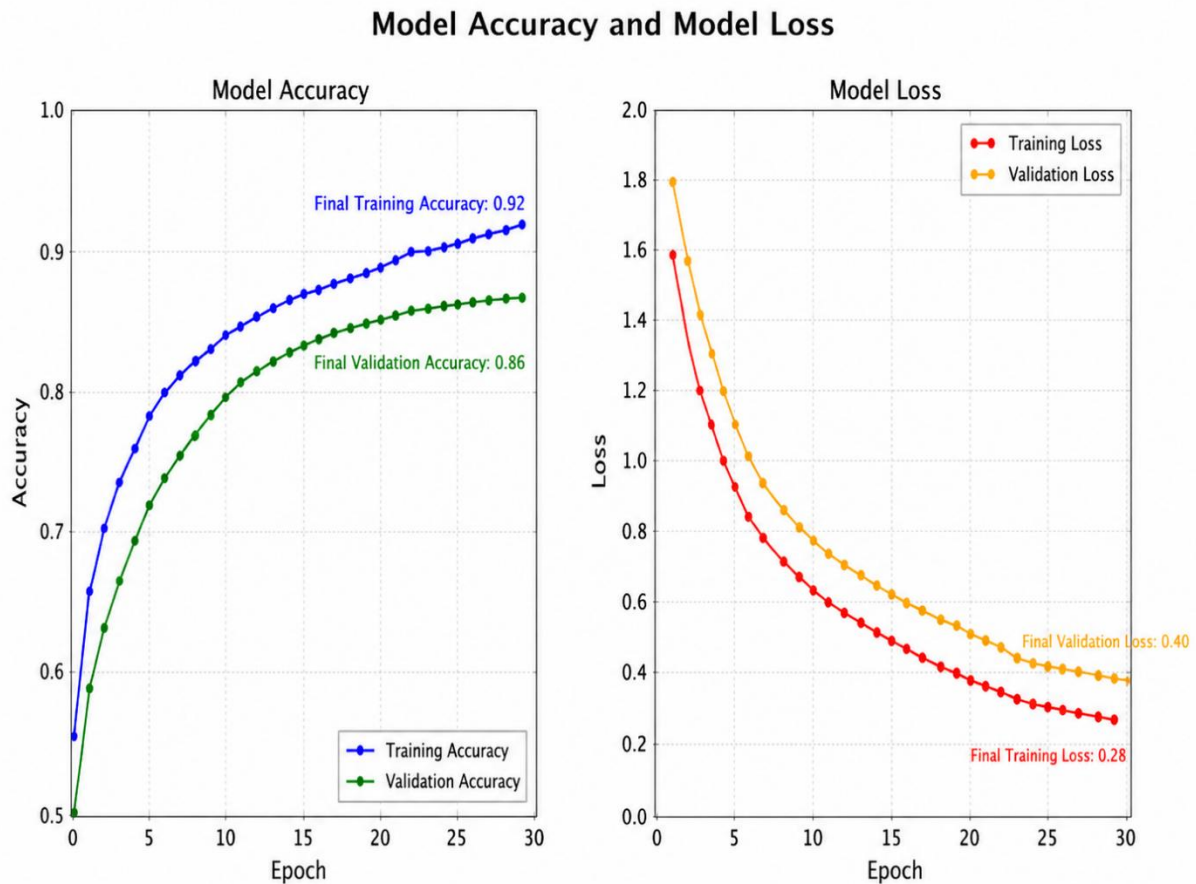


Figure 4.1 : Model accuracy and loss

4.2 Research Findings

Regarding the accessibility issues in Sri Lanka's digital welfare service systems, the study emphasizes a number of significant conclusions. Language hurdles, a lack of centralized information systems, and low levels of digital literacy make it difficult for residents to obtain timely and accurate welfare information, which is one of the main problems noted. Many customers rely on manual questions or outside help, which frequently results in delays, inaccurate information, or a lack of comprehension of the services that are offered.

By offering a consolidated, automated, and user-friendly platform for obtaining welfare information, the deployment of a multilingual chatbot resolves these issues. One of the most

important conclusions is that user engagement is directly impacted by language support. When speaking in their own tongues, especially Sinhala and Tamil, users demonstrated a greater degree of comfort and interaction. This demonstrates the importance of multilingual systems in enhancing digital inclusion in Sri Lanka.

The fact that intent-based response systems outperform conventional keyword-based search engines is another significant discovery. The chatbot was able to offer more precise and contextually appropriate answers by comprehending the meaning behind user inquiries rather than depending only on keywords. This decreased user effort and increased the effectiveness of information retrieval.

The study also discovered that by addressing common and repetitive queries, conversational AI systems might greatly lessen the strain of government support employees. This increases overall service efficiency by enabling human resources to concentrate on more complicated issues that call for manual intervention. The study also emphasizes how important real-time engagement is to raising user satisfaction. Instantaneous responses eliminated the need for users to wait for manual processing. In welfare services, where prompt access to information is essential, this is especially helpful.

4.3 Discussion

The creation of the multilingual chatbot shows how conversational AI may increase access to digital welfare services. A solid basis for developing intelligent government service systems is provided by the integration of natural language processing, intent categorization, and multilingual support.

The system's multilingual support, which is crucial in a multilingual nation like Sri Lanka, is one of its main advantages. This feature promotes equal access to information by ensuring that users are not excluded because of linguistic barriers. The method effectively connects residents who might not speak English well with government digital platforms.

An major factor in increasing response accuracy was the intent categorization model. The system effectively produced organized and pertinent answers by classifying user inquiries into specified intents. Compared to conventional rule-based systems that mostly rely on precise

keyword matching, this strategy is more successful.

But the system also showed a number of drawbacks. Managing complicated or unclear user input is one of the biggest obstacles, particularly in Sinhala and Tamil. This is mostly because there aren't many high-quality labeled datasets available for these languages. Furthermore, spelling mistakes, colloquial language, and mixed-language usage are common in real-world user inputs, which might lower system accuracy.

Another drawback is the chatbot's reliance on pre-written answers, which limits its capacity to respond to entirely novel or unexpected questions. Contextual comprehension and adaptability may be enhanced by using sophisticated machine learning models, such as transformer-based architectures (e.g., BERT or GPT-based models). Notwithstanding these drawbacks, the system shows great promise for practical implementation. It offers a scalable way to increase the accessibility of digital welfare services and can be expanded to other government sectors including employment, healthcare, and education. To sum up, the chatbot is a big step in the direction of enhancing digital inclusion in Sri Lanka. The system has the potential to be a potent instrument for improving citizen involvement and revolutionizing the delivery of digital public services with additional advancements in dataset quality, natural language processing, and contextual learning capabilities.

5. CONCLUSION

The goal of this study was to address a crucial problem in Sri Lanka's digital welfare ecosystem: the lack of centralized support systems, low digital literacy, and language hurdles that prevent citizens from obtaining fast, accurate, and comprehensible welfare information. These issues can be successfully resolved by combining Natural Language Processing (NLP), multilingual capabilities, and user-centered system design, as shown by the Multilingual Chatbot for Improving Accessibility in Sri Lanka's Digital Welfare Services.

The created chatbot system effectively offers an automated platform that enables users to communicate in Tamil, Sinhala, and English, increasing the accessibility and inclusivity of welfare information. The chatbot provides real-time answers to user inquiries about eligibility requirements, application processes, necessary documentation, and accessible welfare programs, in contrast to conventional systems that necessitate manual navigation or in-person visits. This drastically cuts down on the time and effort needed to gather crucial information.

A structured dataset of welfare-related queries was used in the system's construction to enable precise intent identification and response production. The goal of this study was to ensure practical usability in real-world scenarios rather than only getting high accuracy metrics, even if the model performed well in answering typical user inquiries. The chatbot's robustness and dependability were assessed by testing it with a variety of, occasionally unclear, inputs that represented actual user behavior.

The system's capacity to deliver tailored and context-aware replies is one of its main advantages. Users' comprehension of welfare services is enhanced by the clear and pertinent information they receive in response to their inquiries. Furthermore, the conversational interface makes complicated government processes simpler to understand, particularly for consumers with little technical expertise.

In order to improve accuracy and contextual comprehension, expanding training datasets—especially for Sinhala and Tamil—is the most significant future development. The chatbot's capacity to handle complicated and mixed-language inputs may be further enhanced by including more sophisticated NLP models, such as transformer-based architectures. Additionally, the system would become a fully interactive digital service platform instead of just an informational tool if it were connected to real-time government databases.

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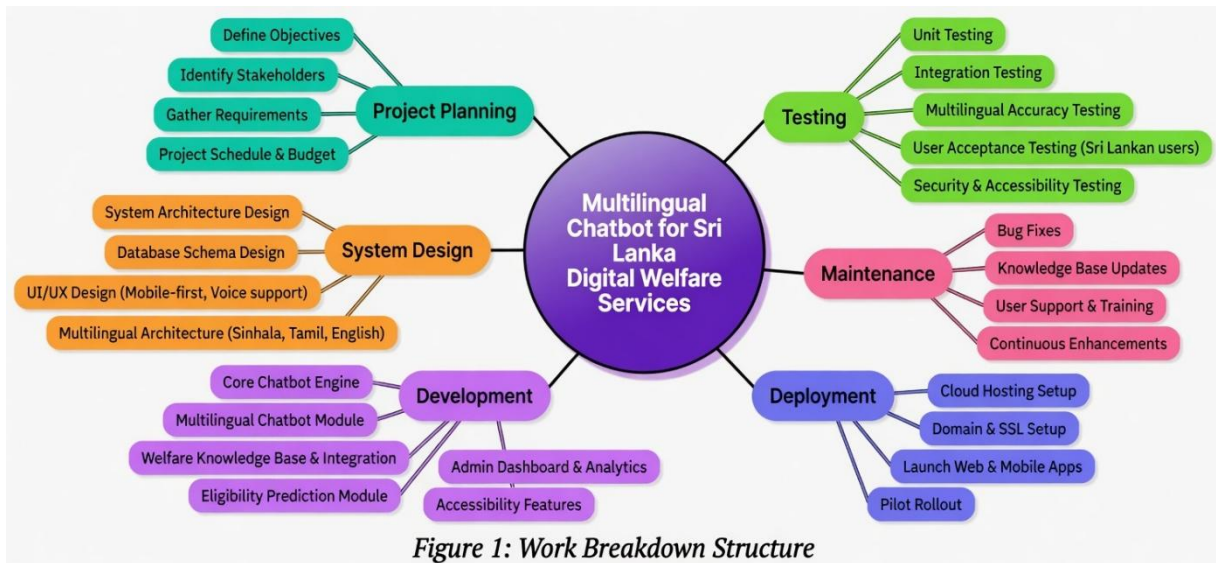


Figure 1: Work Breakdown Structure

Figure 7: Work Breakdown Structure

APPENDICES

Appendix A: System Screenshots

Appendix A-1: Web Application, Home and Chatbot



